

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY**Fifth Semester of B. Tech (ME) Examination****November 2018****ME342/303 Heat and Mass Transfer****Date: 22.11.2018, Thursday****Time: 10.00 a.m. To 01.00 p.m.****Maximum Marks: 70****Instructions:**

1. The question paper comprises two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat figures wherever required.
4. Use of scientific calculator is allowed.

SECTION – I**Q - 1 Answer the question below.**

- (i) Thermal conductivity of air with rise in temperature [01]
 (A) Increases (B) Decreases
 (C) Remain constant (D) May increase or decrease depending on temperature
- (ii) A designer chooses the values of fluid flow rates and specific heats in such a manner [01]
 that the heat capacities of the two fluids are equal. A hot fluid enters the counter flow
 heat exchanger at 100°C and leaves at 60°C. A cold fluid enters the heat exchanger at
 40°C. The mean temperature difference between the two fluids is
 (A) 20°C (B) 40°C
 (C) 60°C (D) 66.7°C
- (iii) Heat conducted through per unit area and unit thick face per unit time when [01]
 temperature difference between opposite faces is unity, is called
 (A) Thermal resistance (B) Thermal coefficient
 (C) Temperature gradient (D) Thermal conductivity
- (iv) Pick up the wrong case. Heat flowing from one side to other depends directly on [01]
 (A) Face area (B) Time
 (C) Thickness (D) Temperature difference
- (v) Define Fin effectiveness and efficiency. [02]
- (vi) In a shell and tube heat exchanger, baffles are provided on the shell side to [01]
 (A) Improve heat transfer (B) Provide support for tubes
 (C) Prevent stagnation of shell side fluid (D) All of these

Q – 2.a What is boiling? What mechanisms are responsible for the very high heat transfer [04]
 coefficients in nucleate boiling?

Q – 2.b Answer any two questions.**[10]**

- (i) Derive the basic differential equation for an extended surface or fins using the principle of energy balance and hence derive the expression for steady state heat transferred by an infinity long fin using the relevant boundary conditions.
- (ii) A furnace walls made up of three layers, one of fire brick, one of insulating brick and one of red brick. The inner and outer surfaces are at 870°C and 40°C respectively. The respective co- efficient of thermal conduciveness of the layer are 1.0, 0.12 and 0.75 W/m-K and thicknesses are 22 cm, 7.5, and 11 cm. assuming close bonding of the layer at their interfaces, find the rate of heat loss per sq. Meter per hour and the interface temperatures.
- (iii) Establish the general differential equation in Cartesian co-ordinates for 3-D unsteady heat conduction by considering an infinitesimal volume element. Deduce there from the conduction equations for unsteady state 2-D flow with heat generation at uniform rate within material and steady 1-D flow without heat generation.

Q - 3 Answer any two questions.**[14]**

- (i) Derive an expression for LMTD for a counter flow recuperative heat exchanger.
- (ii) A spherical container of negligible thickness holding a hot fluid at 140°C and having an outer diameter of 0.4 m is insulated with three layers of each 50 mm thick insulation of $k_1 = 0.02$: $k_2 = 0.06$ and $k_3 = 0.16$ W/m-K. (Starting from inside). The outside surface temperature is 30°C . Determine (i) the heat loss, and (ii) Interface temperatures of insulating layers.
- (iii) A counter flow heat exchanger is employed to cool 0.55 kg/s ($C_p = 2.45$ KJ/Kg $^{\circ}\text{C}$) of oil from 115°C to 40°C by the use of water. The inlet and outlet temperature of cooling water are 15°C and 75°C respectively. The overall heat transfer coefficient is expected to be $1450\text{ W/m}^2\text{ }^{\circ}\text{C}$. Using NTU method, calculate the following: (i) The mass flow rate of water. (ii) The effectiveness of heat exchanger. (iii) The surface area required.

SECTION – II**Q - 4 Answer the question below.**

- (i) Define transmittivity. What is the value of transmittivity of an opaque body? **[02]**
- (ii) Define Solid angle. What is the value of solid angle for hemisphere? **[02]**
- (iii) The ratio of convection to conduction is called _____ number **[01]**
 (A) Nusselt (B) Reynold

(C) Biot (D) Stanton

- (iv) A cube is submerged into a fluid where heat transfer coefficient is h_1 at top surface, h_2 at all the vertical sides and h_3 at the bottom. The average value of heat transfer coefficient will be- [01]

(A) $(h_1+h_2+h_3)/3$ (B) $(h_1+4h_2+h_3)/6$
 (C) $h_1+h_2+h_3$ (D) $h_1 \cdot h_2 \cdot h_3$

- (v) Usual effect of surface roughness on heat transfer is [01]

(A) Increase heat transfer (B) Decrease heat transfer
 (C) Unchanged (D) Depends on size

- Q – 5.a** Define View factor. Derive all possible view factors for equilateral triangular tube geometry. [04]

OR

- Q – 5.a** Derive general equation of mass diffusion in stationary media. [04]

- Q – 5.b** Consider a deep space probe constructed as 1 m diameter polished aluminum sphere. Estimate the equilibrium temperature that the probe reaches if the solar energy received is 300 W/m^2 . For a solar radiation, Absorptivity of aluminum is 0.3 and average emissivity is 0.04. [04]

- Q – 5.c** Three hollow thin walled cylinders having radius 5 cm, 10 cm and 15 cm are arranged concentrically. The temperatures of the innermost and outermost cylindrical surfaces are 1000 K and 300 K respectively. Assuming vacuum between the annular spaces, determine the steady state temperature attained by the cylindrical surface having radius of 10 cm. Proceed to calculate the heat flow per m^2 area of cylinder with radius 5 cm. [06]

OR

- Q – 5.c** Two large parallel plates with emissivity 0.4 are maintained at different temperatures and exchange heat only by radiation. What is percentage change in net radiation heat transfer would occur if two equally large radiation shields with surface emissivity 0.04 are introduced in parallel to the plates? [06]

- Q – 6.a** Consider laminar flow over a flat plate. Will the heat transfer coefficient change with distance from the leading edge? How about the friction coefficient? [03]

- (i) distance from the leading edge? How about the friction coefficient?
 (ii) From first principles, deduce the following for forced convection $Nu = f(Re, Pr)$ [04]
 Notations bear the usual meaning.

- Q – 6.b** A long 10-cm-diameter steam pipe whose external surface temperature is 110°C passes through some open area that is not protected against the winds. Determine the rate of heat loss from the pipe per unit of its length when the air is at 1 atm pressure and 10°C and the wind is blowing across the pipe at a velocity of 8 m/s. The properties of air at the average film temperature - $k = 0.02808 \text{ W/m-K}$; $\text{Pr} = 0.7202$; $\nu = 1.896 \times 10^{-5} \text{ m}^2/\text{s}$ **[07]**

OR

- Q – 6.a** In which mode of heat transfer is the convection heat transfer coefficient usually **[03]**
- (i) higher, natural convection or forced convection? Why?
 - (ii) Physically, what does the Grashof number represent? How does the Grashof number differ from the Reynolds number? **[04]**
- Q – 6.b** Consider flow of water entering in a tube with velocity 3 m/s and 10°C which is to be heated up to a temperature of 80°C . The tube diameter is 2cm and equipped with an electric resistance heater on its surface. Considering case of internal flow, calculate the power rating of the heater and the inner surface temperature. Various properties at the mean temperature are- **[07]**

$k = 0.637 \text{ W/m.-K}$; $\text{Pr} = 3.91$; $\nu = 0.602 \times 10^{-6} \text{ m}^2/\text{s}$; $\rho = 990.1 \text{ kg/m}^3$; $C_p = 4180 \text{ J/kg-K}$
